



Department of Mechanical Engineering
Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. Mechanical Engineering

Course Code & Title: ME8693 Heat and Mass Transfer

Name of the Faculty member(s): Mr.M.Ashok Kumar, AP (SG)/Mechanical

Innovative Practice Description

Unit / Topic: Unit – I to Unit – V – All Tutorial classes

Course Outcome:

CO1 – Apply heat conduction equations to different surface configurations under steady state and transient conditions and solve problems

CO2 – Apply free and forced convective heat transfer correlations to internal and external flows through/over various surface configurations and solve problems

CO3 – Explain the phenomena of boiling and condensation, apply LMTD and NTU methods of thermal analysis to different types of heat exchanger configurations and solve problems.

CO4 – Compare cogeneration's performance with conventional power generation and various heat recovery method of cogeneration.

CO5 – Investigate the performance of Refrigeration and Air-conditioning system under various working and load conditions.

Topic Learning Outcome:

TLO 3: Apply the one dimensional heat conduction (with and without heat generation) in Plane wall, Hollow cylinder, Hollow sphere and extended surfaces.

TLO 7: Compute the heat transfer efficient and convection heat transfer during external flow over plates and cylinder and sphere.

TLO 8: Calculate the heat transfer coefficient and convection heat transfer through internal flow through tubes and natural convection.

TLO 11: Determine the heat transfer coefficient during boiling and condensation.

TLO 13: Calculate the overall heat transfer coefficient and fouling factor in heat exchanger and identify the suitability of NTU and LMTD approach.

TLO 16: Calculate the radiant heat exchange between radiation shield and able to apply the electrical analogy concept.

TLO 20: Determine the mass transfer coefficient during steady state molecular diffusion.

TLO 21: Calculate the convective mass transfer rate.

Activity Chosen: Peer learning/collaborative learning

Justification:

- The students who are hesitate to clarify the doubt with faculty members shall clarify their doubt with their friends and improve the ability of problem solving skill.
- It creates the interest to solve the problem among the students while discussing with their friends at the time of solving the problems through peer learning.

- **Time Allotted for the Activity:** 50 min x 30 tutorial classes

Details of the Implementation:

The problem solving skill is very important to the students to improve their knowledge in the relevant engineering field and it will also help the students to perform well in the exam and also in the industry. The problem solving skills of the students can be improved by the peer learning / collaborative learning activity, so it is planned to conduct all the tutorial classes by peer learning / collaborative learning.

Problem Statement

Annexure – I – Tutorial sheets and hand out material of problems that contains all five units

PO / PSO mapping for the activity:

Innovative Practice	PO1	PO2	PO9	PSO3
Level of mapping	3	3	1	3
Justification for correlation	To solve the problem the student will apply the mathematical, science and engineering fundamentals – Level 3	Derive the formula using mathematics, natural science and engineering science to calculate the performance parameters – Level 3	The tutorial hours are conducted by the concept of peer learning at that time the student's individual and team work is improved – Level 1	Calculate the performance parameters steam nozzle, boilers, steam turbine, cogeneration plant and Refrigeration and Air-conditioning systems – Level 3
	(1 – Low	2 – Moderate	3 – High)	

CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO9	PSO3
CO1	3	3	1	1	3
CO2	3	3	1	1	3
CO3	3	3	1	1	3
CO4	3	3	1	1	3
CO5	3	3	1	1	3
	(1 – Low	2 – Moderate	3 – High)		

Reflective Critique:

- **Feedback of practice from students and other stakeholders:** (samples to be enclosed)

Feedback received from students and the sample is enclosed

- **Benefit of the practice:**

- The problem solving skill of the students is improved
- Create interest among the bright students to solve the advanced problems given in the hand out.

- **Whether the practice is adopted in any of the courses early: Yes**

(If yes provide the details and the modifications you have adopted)

The tutorial is conducted for the entire unit using the peer learning/collaborative learning concept. The complete problems for all of the units are given in the first class itself to encourage the bright students to show interest in solving the advance problems on their own from the hand note before solving / discussing in the class.

Challenges faced in implementation: Nil

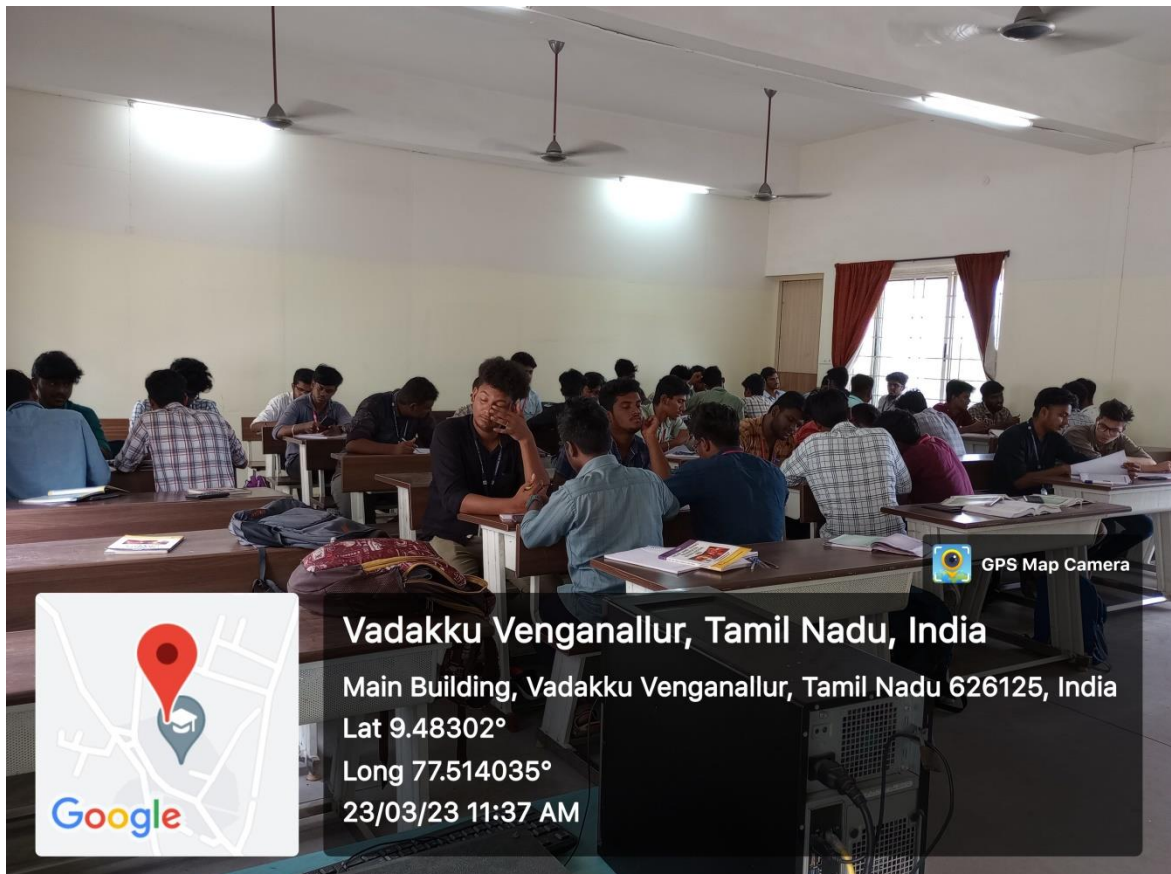
References:

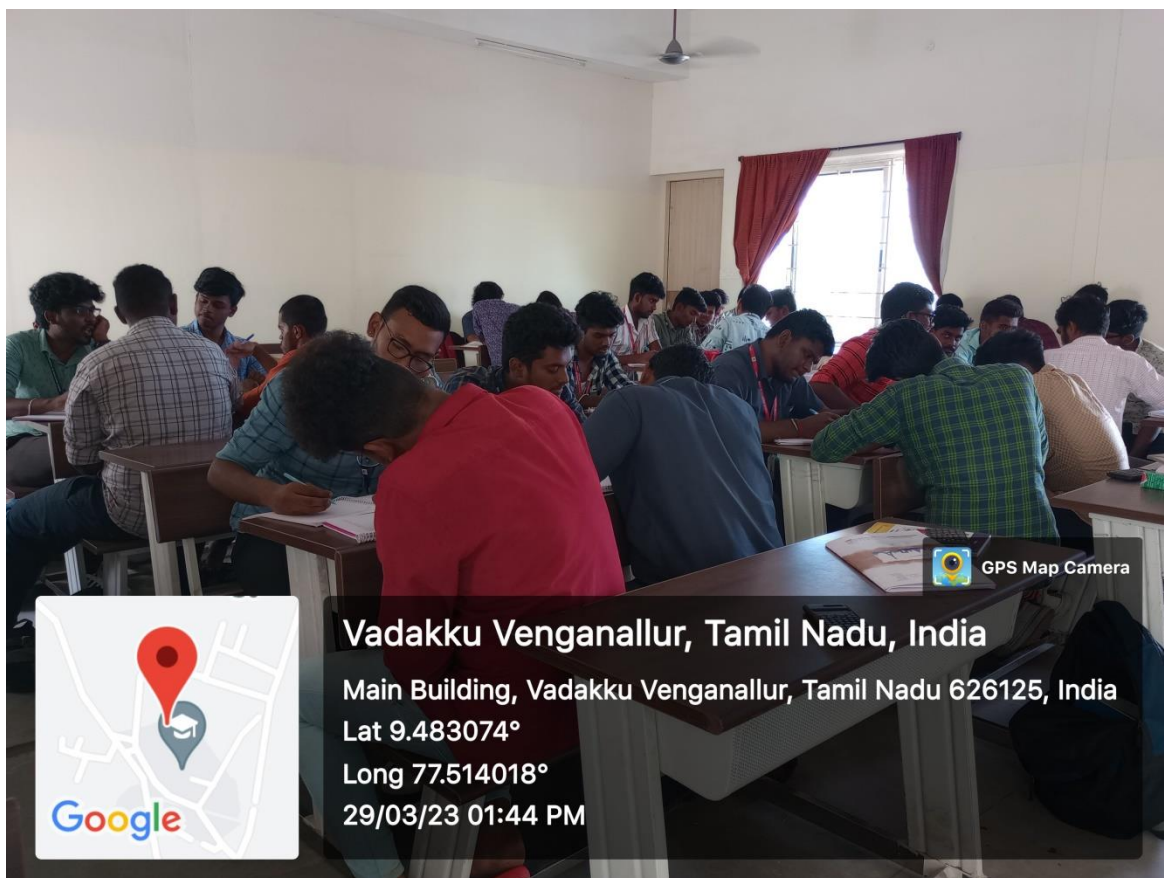
- Yunus A. Cengel, “Heat Transfer – A Practical Approach”, Tata McGraw Hill, 5th Edition 2015.
- Frank P. Incropera and David P. Dewitt, “Fundamentals of Heat and Mass Transfer” John Wiley & Sons 2007.

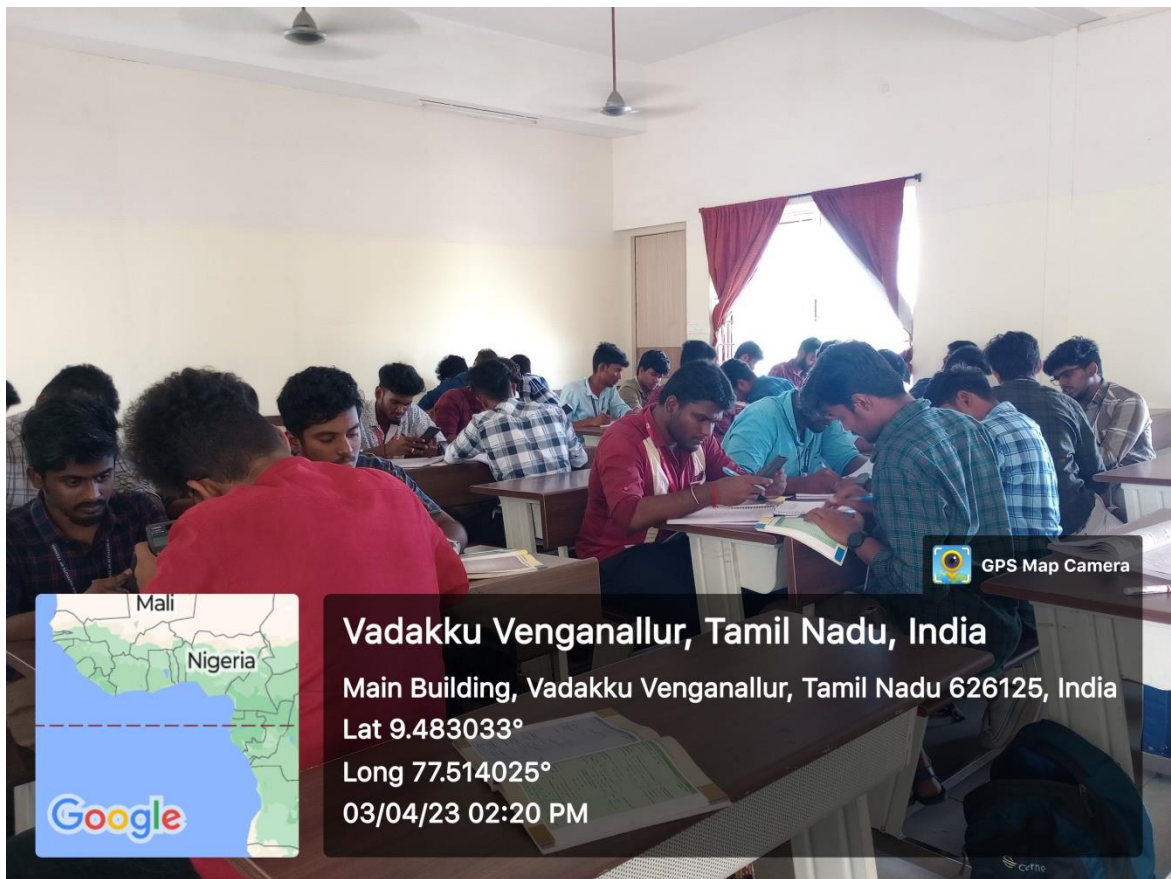
Images / Screenshot of the practice:













Signature of Faculty Member

HOD